



Modelling Financial Markets

Chapter 10: Market Microstructure



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CARA: Constant Absolute Risk Aversion
- ▣ CEX: Centralised EXchange
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ CI: Confidence Interval
- ▣ DEX: Decentralised EXchange
- ▣ EDGE: Efficient Discrete Generalized Estimator (of the bid-ask spread)
- ▣ EKOP: Easley–Kiefer–O'Hara–Paperman (model of informed trading)
- ▣ ETF: Exchange-Traded Fund
- ▣ FOK: Fill-Or-Kill (order)
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GMM: Generalized Method of Moments
- ▣ GTC: Good-Till-Cancelled (order)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HFT: High-Frequency Trading
- ▣ IOC: Immediate-Or-Cancel (order)
- ▣ LLM: Large Language Model
- ▣ LOB: Limit Order Book
- ▣ MA: Moving Average
- ▣ MRR: Madhavan–Richardson–Roomans (structural model of trading costs)

Acronyms used in this material (2/2)

- ▣ NYSE: New York Stock Exchange
- ▣ OHLC: Open, High, Low, Close
- ▣ OLS: Ordinary Least Squares
- ▣ PIN: Probability of INformed trading
- ▣ SEC: Securities and Exchange Commission
- ▣ TWAP: Time-Weighted Average Price
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VAR: Vector AutoRegression
- ▣ VECM: Vector Error-Correction Model
- ▣ VIX: CBOE Volatility Index
- ▣ VPIN: Volume-synchronised Probability of INformed trading
- ▣ VWAP: Volume-Weighted Average Price
- ▣ XRP: XRP (the native token of the XRP Ledger)

Outline

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Crypto Market Microstructure

Liquidity at the Bucharest Stock Exchange

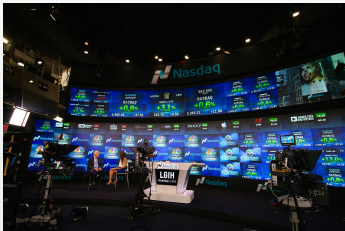
Case Study: Quantifying the HFT Arms Race

AI for Scientific Discovery

Summary

References

From the Trading Floor to the Matching Engine



Nasdaq MarketSite, Times Square, New York: a studio,
not a trading floor

Photo: Luca Marfè (2015); CC BY 2.0; Wikimedia Commons

- For two centuries prices were made by people in a room: brokers, specialists, open outcry
 - ▶ The specialist kept the order book on paper and quoted a bid and an ask
- Today almost all equity, futures and crypto trading runs on computers that match orders in microseconds
- The rules of the matching engine (priority, tick size, auctions) shape prices and costs [Gould et al. \(2013\)](#)
- This chapter: price formation, trading costs and the econometric methods used to study them
 - ▶ Textbooks: [Hasbrouck \(2007\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#), [Franke, Härdle & Hafner \(2019\)](#)

Data Used in This Chapter

Series	Content	Period
SPY, 5-minute bars	ETF on the S&P 500; regular session 09:30–16:00 New York time; 1 477 full days	2020–2026
Bitcoin, 5-minute bars	24 hours a day, 7 days a week; 208 596 bars	2024–2026
US stocks, daily	SPY, Apple, Microsoft, Nvidia, JPMorgan, GameStop, Strategy	2016–2026
BVB stocks, daily	Ten blue chips and the BET ETF; values in lei converted to USD at the BNR reference rate	2016–2026
Crypto-assets, daily	Bitcoin, Ethereum, Solana, XRP, Dogecoin, Cardano, Litecoin	2016–2026
VIX, daily	Implied volatility of the S&P 500	2020–2026

- Daily data: open, high, low, close and volume; prices adjusted for dividends and splits; days without trades removed
- Order books in Section 2 and the bid-ask bounce in Section 3 are **simulated**, and labelled as such

Liquidity and the Limit Order Book: Definitions

- **Liquidity**: trading a quantity quickly at a price close to fair value; no single number captures it
 - ▶ **Tightness** (the spread), **depth** (size needed to move the price), **resiliency** (speed of recovery) Kyle (1985); **immediacy** Harris (2003)
- **Limit order book** (LOB): all standing limit orders, by price and time of arrival Gould et al. (2013)
- Best bid $b_t <$ best ask a_t ; mid-price $m_t = (a_t + b_t)/2$; relative spread $(a_t - b_t)/m_t$ in basis points (1 bp = 0.01%)
 - ▶ **Depth**: quantity waiting at a price level; **tick size**: smallest price step (USD 0.01 for most US stocks above USD 1)

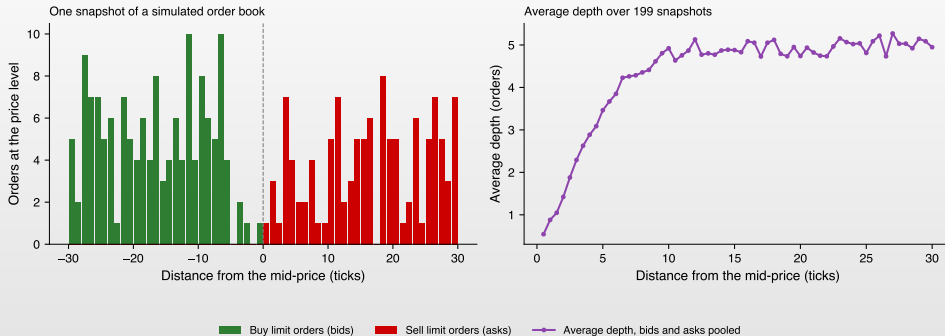
Orders, Priority and Matching

- **Market order**: immediate execution against the available liquidity (within venue rules), uncertain price; **limit order**: price bounded by the limit, uncertain execution
 - ▶ Variants: IOC, FOK, GTC (glossary); **stop** order: becomes a market order once a trigger price trades; **iceberg** order: displays only part of its size, the rest waits hidden in reserve
 - ▶ Resting (non-marketable) limit orders supply liquidity; market orders and marketable limit orders (a buy limit at or above the best ask) consume it
- **Price-time priority**: the better price first, then the earlier order at the same price (a queue); queue position is valuable (Section 8)
- **Call auctions** at the open and close: orders are pooled and matched at the single price that maximises the executed quantity
- A market buy of size Q walks up the ask levels: it pays the half-spread plus the cost of the deeper levels (the impact function below)

A Simple Model of the Order Book

- **Zero-intelligence** model: orders arrive at random, with no strategy [Farmer, Patelli & Zovko \(2005\)](#)
 - ▶ Limit orders: rate α per price level, placed uniformly up to 40 ticks from the opposite best quote
 - ▶ Market orders: rate μ per side; cancellations: rate δ per standing order
- Here $\alpha = 0.1$, $\mu = 0.5$, $\delta = 0.02$; 300 000 events, unit orders
- Farmer, Patelli and Zovko show that such a model already predicts much of the cross-section of spreads on the London Stock Exchange
- Use: a laboratory where spread, depth and impact can be measured exactly

A Simulated Order Book

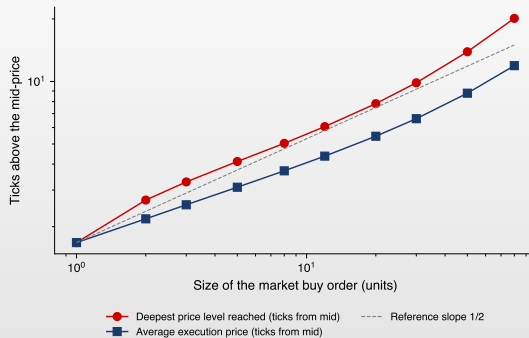


Left: one snapshot (bids green, asks red); right: average depth by distance from the mid, bids and asks pooled

Reading the Simulated Order Book

- Mean spread 3.23 ticks, median 3; it exceeds 6 ticks 10% of the time
- Depth is thin near the mid and grows away from it
 - ▶ 0.8 orders at 1 tick, 4.8 at 10 ticks, 5.0 at 25 ticks (average over 199 snapshots)
- Reason: market orders eat the best levels first, so liquidity near the price is scarce [Tóth et al. \(2011\)](#)
- Real order books show the same rise away from the mid, often with a maximum a few ticks away [Bouchaud, Bonart, Donier & Gould \(2018\)](#)

The Price Impact of a Market Order



□ Market buy orders of 1 to 80 units sent against each simulated snapshot; log-log axes

Reading the Impact Function

- ▣ A 1-unit order reaches 1.7 ticks above the mid; an 80-unit order reaches 20.1 ticks
- ▣ **Concave** impact: slope 0.53 on log-log axes, close to $1/2$
 - ▶ Average execution price: slope 0.44; 80 units cost 11.9 ticks on average
- ▣ Doubling the order does not double the cost: depth grows with distance from the mid
- ▣ Real markets show a similar concave law for large orders split over time (Section 7)

Why Is There a Spread?

- A liquidity provider buys at the bid and sells at the ask; the spread pays three costs [Huang & Stoll \(1997\)](#)
- **Order processing:** fees, technology, clearing
- **Inventory:** holding an unwanted position exposes the provider to price risk
- **Adverse selection:** some counterparties know more [Bagehot \(1971\)](#)
 - ▶ The provider incurs losses when trading with informed traders and must recover them from uninformed traders (Section 6)

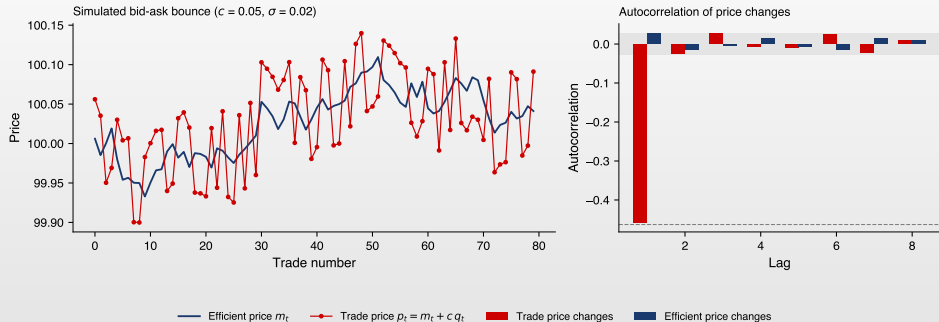
Quoted, Effective and Realised Spread

- ▣ Trade at price p_t , direction $q_t = +1$ (buyer-initiated) or -1 (seller-initiated), mid-price m_t
- ▣ **Quoted:** $S_t^Q = (a_t - b_t)/m_t$; **effective:** $S_t^E = 2q_t(p_t - m_t)/m_t$, twice the signed distance from the mid (round-trip equivalent; the one-way cost is half)
- ▣ **Realised:** $S_t^R = 2q_t(p_t - m_{t+\Delta})/m_t$, the provider's revenue marked to the mid after Δ (e.g. 5 minutes), before fees and inventory costs
 - ▶ Price impact $= S_t^E - S_t^R = 2q_t(m_{t+\Delta} - m_t)/m_t$: the lasting move of the mid after the trade, usually attributed to adverse selection, though inventory and order-flow pressure also contribute
- ▣ Quoted spread: quotes only; effective and realised spread: trades, the prevailing quotes and the trade direction, signed for example by the rule of [Lee & Ready \(1991\)](#)
- ▣ Without such data we **estimate** the spread from prices only: the next slides

The Roll Model

- Efficient price: random walk $m_t = m_{t-1} + \varepsilon_t$, $\varepsilon_t \sim (0, \sigma^2)$ i.i.d. [Roll \(1984\)](#)
 - ▶ Trade price $p_t = m_t + c q_t$, with half-spread c and $q_t = \pm 1$ with probability $1/2$, i.i.d. over time and independent of ε
- Price change: $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$
- $\text{Var}(\Delta p_t) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$
- **Roll estimator:** $\hat{s} = 2\hat{c} = 2\sqrt{-\widehat{\text{Cov}}(\Delta p_t, \Delta p_{t-1})}$
 - ▶ The bid-ask bounce creates negative autocorrelation even when the efficient price is a random walk

The Bid-Ask Bounce (Simulated)



- 5 000 simulated trades with $c = 0.05$ and $\sigma = 0.02$; grey band: $\pm 1.96/\sqrt{n}$; dashed line: the theoretical first-order autocorrelation

Reading the Roll Simulation

- True spread $2c = 0.10$; Roll estimate from the simulated trades: $\hat{s} = 0.0998$
- First-order autocorrelation -0.458 , theory $-c^2/(\sigma^2 + 2c^2) = -0.463$
- No autocorrelation at lags 2 and beyond: the bounce lasts only one trade
- The efficient price changes are uncorrelated: the market is efficient, yet trade prices are not a random walk

When the Roll Estimator Fails

- With real data the sample covariance is often **positive**: $\hat{s}$ does not exist
 - ▶ SPY, 5-minute bars: the covariance within the day is positive on 41% of days
- Causes: momentum or news trends, orders split into many trades, stale prices, bars instead of trades
- With daily data the bounce (a few bp) is buried under the daily volatility (103 bp for SPY over the last two years)
- Hence estimators based on the daily high and low: the next two slides

Statistical Properties of the Roll Estimator

- ▣ $\hat{\gamma}_1$: sample autocovariance of Δp from T price changes (demeaned); $\gamma_0 = \sigma^2 + 2c^2$, $\gamma_1 = -c^2$ Harris (1990)
 - Bias from demeaning: $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$, negative even when $c = 0$
- ▣ Gaussian approximation (Bartlett's formula for an MA(1)): $T \text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$
- ▣ Hence, ignoring the demeaning bias, $P(\hat{\gamma}_1 > 0) \approx \Phi(\sqrt{T} \gamma_1 / \sqrt{\gamma_0^2 + 3\gamma_1^2}) \approx \Phi(\sqrt{T} \gamma_1 / \gamma_0)$ when $c^2 \ll \sigma^2$: close to 1/2 whenever $c^2 \ll \sigma^2 / \sqrt{T}$
- ▣ Delta method: $\text{se}(\hat{s}) \approx \sqrt{(\gamma_0^2 + 3\gamma_1^2)/T} / c$; $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$ is also biased (Jensen's inequality, truncation at 0)
- ▣ Seminar A1–A2 derive these moments and the standard error step by step

SPY: Do Positive Covariances Reject the Roll Model?

- ▣ Per day, 77 five-minute returns; a one-tick spread means a half-spread $c = 0.10$ bp, against a bar standard deviation of 9.2 bp ($c/\sigma = 0.011$)
- ▣ Monte Carlo under the Roll model with a one-tick spread (200 samples of the same size): $\hat{\gamma}_1 > 0$ on 45.3% of days, 95% band [43.0, 48.1]%
 - ▶ Observed: 40.8%, below the band: the data are **more** negatively autocorrelated than a one-tick bounce, not less
- ▣ With no spread at all ($c = 0$) the pooled Roll estimate is still 2.06 bp [1.50, 2.53]: most of the observed 2.82 bp is demeaning bias
- ▣ Bias-corrected: 1.92 bp, still about ten ticks: the remaining negative autocorrelation must come from sources other than the bounce, such as transient price pressure

A Structural Model of Trading Costs: MRR

- ▣ Roll assumes i.i.d. trade signs; with first-order Markov signs, $\text{Corr}(q_t, q_{t-k}) = \rho^k$, independent of ε and without information, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2(1 - \rho)^2$: Roll underestimates c
- ▣ MRR model [Madhavan, Richardson & Roomans \(1997\)](#): $p_t = m_t + \phi q_t + \xi_t$,
 $m_t = m_{t-1} + \theta (q_t - \mathbb{E}[q_t | q_{t-1}]) + \varepsilon_t$
 - ▶ θ : permanent (adverse-selection) impact of the order-flow **surprise**; ϕ : transitory cost (processing, inventory)
 - ▶ q_t a first-order Markov chain; public news ε_t and rounding noise ξ_t i.i.d., mean zero, independent of each other and of q
- ▣ With $u_t = \Delta p_t - (\phi + \theta)q_t + (\phi + \rho\theta)q_{t-1}$: GMM on $\mathbb{E}[q_t q_{t-1} - \rho q_{t-1}^2] = 0$, $\mathbb{E}[u_t - a] = 0$, $\mathbb{E}[(u_t - a)q_t] = 0$, $\mathbb{E}[(u_t - a)q_{t-1}] = 0$, a the drift (constant) of u_t
- ▣ Implied effective spread $2(\phi + \theta)$; adverse-selection share $\theta/(\phi + \theta)$
- ▣ Needs signed trades (trade-and-quote data); our public 5-minute bars do not identify q_t , so MRR is not estimated here

The Corwin–Schultz Estimator

- Idea: the daily high is usually a buy at the ask, the low a sell at the bid; the high-low range = volatility + spread [Corwin & Schultz \(2012\)](#)
- The volatility part grows with the length of the interval, the spread part does not: compare one day with two days
- $\beta = \sum_{j=0}^1 \left[\ln \frac{H_{t+j}}{L_{t+j}} \right]^2$, $\gamma = \left[\ln \frac{H_{t,t+1}}{L_{t,t+1}} \right]^2$, $\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}}$
- $\hat{S} = \frac{2(e^\alpha - 1)}{1 + e^\alpha}$, with $H_{t,t+1}$, $L_{t,t+1}$ the high and low over the two days
 - ▶ Negative two-day estimates are set to 0, then averaged over the period
 - ▶ The derivation assumes no price move overnight; for daily data the paper shifts H_{t+1} , L_{t+1} down by $L_{t+1} - C_t$ if $L_{t+1} > C_t$, up by $C_t - H_{t+1}$ if $H_{t+1} < C_t$ (C_t the close)

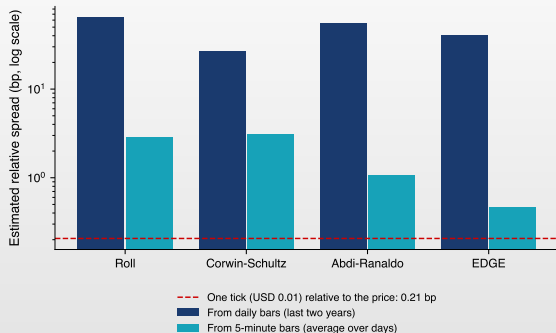
The Abdi–Ranaldo Estimator

- ▣ Uses the close c_t and the mid-range $\eta_t = (\ln H_t + \ln L_t)/2$ as a proxy for the mid-price [Abdi & Ranaldo \(2017\)](#)
- ▣ $\hat{S}^2 = 4 \mathbb{E} [(c_t - \eta_t)(c_t - \eta_{t+1})]$, with c_t the log close
- ▣ The close sits at the bid or the ask; the distance to the neighbouring mid-ranges reveals the half-spread
- ▣ If the sample mean is negative, the estimate is set to 0
- ▣ For crypto-assets, both high-low estimators track the time variation of high-frequency liquidity well [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)
 - ▶ The same study finds the Amihud ratio better for **levels** and for comparing venues

The EDGE Estimator

- ▣ EDGE (Efficient Discrete Generalized Estimator) uses all four prices: open, high, low and close [Ardia, Guidotti & Kroencke \(2024\)](#)
- ▣ Two moment conditions identify S^2 from products of log-returns between the open, the close and the mid-range η_t
 - ▶ Each gives an unbiased estimate $\hat{e}_k$ with variance v_k ; if the two are treated as uncorrelated (diagonal weighting), the minimum-variance combination is $\hat{S}^2 = (v_2 \hat{e}_1 + v_1 \hat{e}_2) / (v_1 + v_2)$
- ▣ Unbiased for S^2 , with minimum variance among estimators built from these moments, under discrete observation of the efficient price; $\hat{S} = \sqrt{\hat{S}^2}$, truncated at 0, is not exactly unbiased for S
- ▣ Averaging over windows: the authors recommend signed estimates with negative values reset to zero, since more negative estimates tend to go with larger spreads [Ardia, Guidotti & Kroencke \(2024\)](#)

Spread Estimators at Two Frequencies: SPY

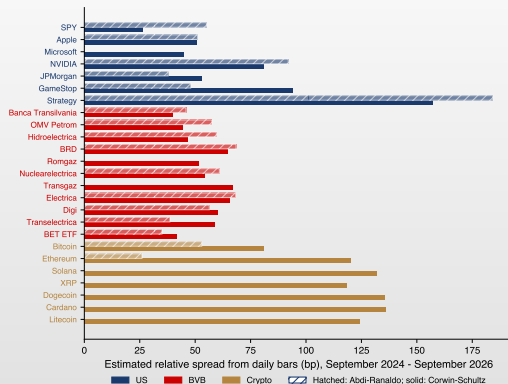


- Daily bars: September 2024 – September 2026; 5-minute bars, 1 477 days (2020–2026): Roll and Abdi–Rinaldo from the daily moments averaged over days, then the square root; Corwin–Schultz and EDGE (negatives set to 0) averaged over days; the two windows differ, so the contrast mixes frequency and sample period

Reading the Frequency Chart

- The **quoted** spread cannot be smaller than one price step (USD 0.01, i.e. 0.21 bp for SPY); an effective spread can, through price improvement
- From daily bars: Roll 64.3 bp, Corwin–Schultz 26.5 bp, Abdi–Ranaldo 55.1 bp
 - ▶ Two orders of magnitude above one tick for one of the most traded securities: implausible as a cost, consistent with volatility dominating the daily range [Jahan-Parvar & Zikes \(2023\)](#)
- From 5-minute bars: Roll 2.82, Corwin–Schultz 3.11, Abdi–Ranaldo 1.06 bp: closer, still above the tick
 - ▶ EDGE: 40.1 bp from daily bars, but 0.46 bp from 5-minute bars (95% CI resampling days [0.42, 0.49]): the closest to one tick
 - ▶ One tick is a plausibility benchmark, not a validation: that needs effective spreads from matched trades and quotes

Spread Estimates Across 25 Assets



□ Daily bars, September 2024 – September 2026; BVB stocks in lei, prices adjusted for dividends and splits

Reading the Cross-Section of Spread Estimates

- Median Corwin–Schultz estimate: US 53 bp, BVB 54 bp, crypto 124 bp
 - ▶ BVB blue chips look as liquid as US large caps: implausible, since they trade a thousand times less value (next section)
- The ranking follows volatility: crypto-assets and the most volatile stocks get the widest estimates
- Abdi–Rinaldo is 0 for 8 of the 25 assets: a negative sample mean
- Levels need validation against effective spreads [Jahan-Parvar & Zikes \(2023\)](#); for crypto-assets, [Brauneis, Mestel, Riordan & Theissen \(2021\)](#) find these estimators reliable for changes over time rather than for levels

Recap: Order Book and Spreads

- Even random order flow creates a spread, a depth profile that rises away from the mid and concave impact [Farmer, Patelli & Zovko \(2005\)](#)
- The spread pays for order processing, inventory risk and adverse selection
- Bid-ask bounce makes successive transaction-price changes negatively autocorrelated: the Roll estimator
- In small samples the Roll covariance is biased and often positive even when the model is true [Harris \(1990\)](#)
- High-low estimators work at high frequency; on daily data they are dominated by volatility when the spread is small relative to it [Jahan-Parvar & Zikes \(2023\)](#); EDGE is the efficient OHLC estimator

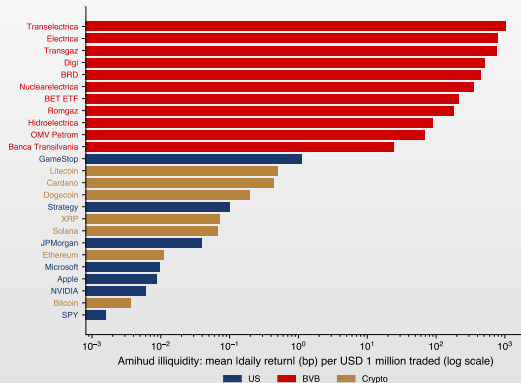
The Amihud Illiquidity Ratio

- Daily price impact per unit of money traded [Amihud \(2002\)](#)

►
$$\text{ILLIQ}_i = \frac{1}{D_i} \sum_{d=1}^{D_i} \frac{|r_{i,d}|}{\text{DVOL}_{i,d}}, \text{ with } r_{i,d} \text{ the daily return and } \text{DVOL}_{i,d} \text{ the traded value}$$

- Units here: basis points of price move per USD 1 million traded
- Needs only daily returns and volumes: available for every market and every period
- A low-frequency proxy of the Kyle price-impact coefficient (Section 6) [Hasbrouck \(2009\)](#)
- Traded value of BVB stocks: value in lei divided by the BNR reference rate of the day; crypto: traded value in USD summed across exchanges

Amihud Illiquidity Across 25 Assets

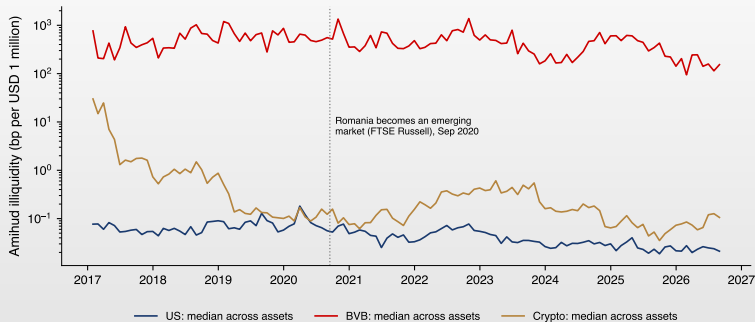


September 2024 – September 2026; log scale; each bar is the mean over about 500 trading days (730 for crypto)

Reading the Amihud Ratios

- ▣ SPY: 0.0016 bp per USD 1 million; median daily value USD 40 124 million
 - ▶ Bitcoin: 0.0037 bp, with USD 41 834 million a day across exchanges
- ▣ Banca Transilvania, the most liquid BVB stock: 24.7 bp; median value USD 3.90 million a day
 - ▶ About 15 541 times the SPY figure; Transelectrica: 1024.0 bp
- ▣ Three clusters: US large caps and the top crypto-assets, then smaller crypto-assets and GameStop, then the BVB
- ▣ Unlike the spread estimators, the Amihud ratio separates markets by orders of magnitude

Illiquidity over Time: Three Markets

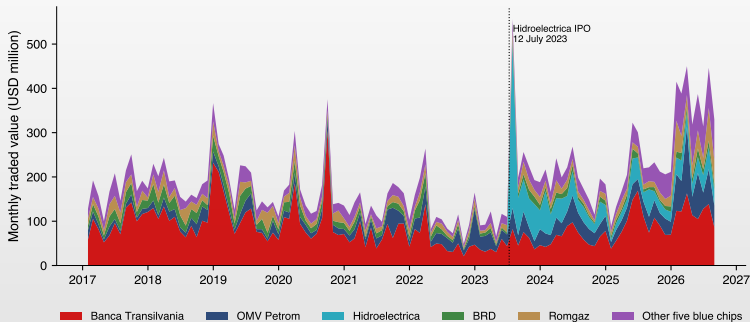


□ Monthly Amihud ratio, median across the assets of each group (without SPY and the BET ETF); log scale

Reading Illiquidity over Time

- ▣ Crypto: from 7.6 bp per USD 1 million in 2017 to 0.089 in 2026
 - ▶ Crypto trading grew from a niche to institutional volumes; spot ETFs followed (Chapter 16)
- ▣ BVB: from 435 to 169: more liquid, but still about 7 212 times the US median
- ▣ US: from 0.062 to 0.023: large caps became even cheaper to trade
- ▣ Romania moved from frontier to emerging-market status in September 2020 [BVB \(history\)](#); the BVB median shows no break at that date

Traded Value of BVB Blue Chips



□ Monthly sum of daily traded value, ten blue chips, converted to USD at the BNR reference rate

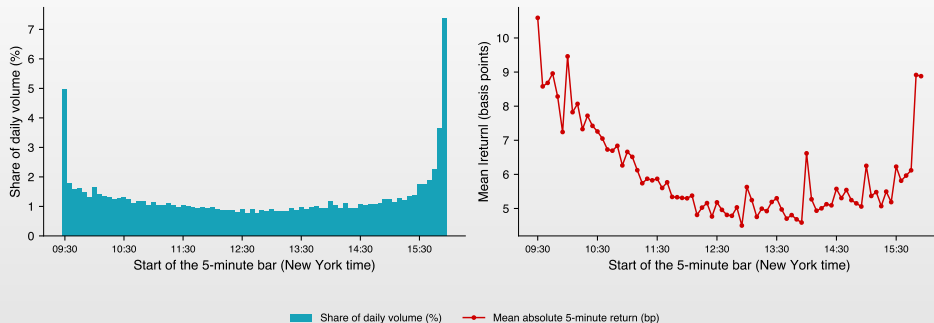
Reading the BVB Traded Value

- ▣ Ten blue chips: USD 2 098 million traded in 2017, 1 494 million in 2022, 2 520 million in 2025
- ▣ Concentration: Banca Transilvania alone made 44% of the 2025 value, Hidroelectrica 13%
 - ▶ The Hidroelectrica listing of July 2023 added a second large, liquid stock
- ▣ One day of SPY trading exceeds a whole year of the ten BVB blue chips
- ▣ Consequence for a fund: a position of a few million USD takes days to build or unwind

Is Liquidity Priced?

- **Level:** investors demand a higher return for assets that are costly to trade [Amihud & Mendelson \(1986\)](#)
 - ▶ Expected return increases with the spread, and concavely: long-horizon investors hold the illiquid assets
- **Time series:** expected market illiquidity raises future returns; an unexpected rise lowers current prices [Amihud \(2002\)](#)
- **Liquidity risk:** stocks that fall when market liquidity dries up earn a premium [Pástor & Stambaugh \(2003\)](#), [Acharya & Pedersen \(2005\)](#)
- Seminar Part C tests the time-series prediction on the BVB, US and crypto markets

SPY Within the Trading Day



- 78 five-minute bars per day, 1 477 full days (October 2020 – September 2026); the first bar's return starts at the opening price

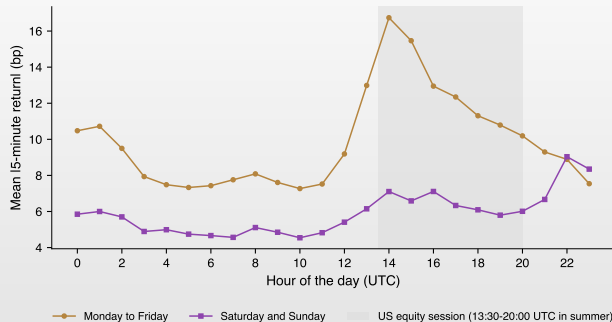
Reading the U-Shape

- ▣ Volume: the first bar carries 5.0% of the day, a midday bar 0.9%; the last 30 minutes 18.6% (95% CI resampling days [18.3, 19.0]) against 7.7% if even
- ▣ Mean $|r|$: 10.6 bp in the first bar against 5.1 bp at midday; ratio 2.06, 95% CI [1.98, 2.15]: a systematic pattern, not a few extreme days
- ▣ Resample whole days: bars of one day share its volatility (seminar B1); spikes at 10:00 and 14:00 are scheduled releases

Why a U-Shape?

- Documented for NYSE stocks by [Wood, McInish & Ord \(1985\)](#): returns and volume are highest at the open and the close
- **Open**: overnight news is absorbed; uncertainty is highest
 - ▶ Informed traders act early, before the information spreads
- **Clustering**: uninformed traders who can choose the time to trade together, and liquidity attracts liquidity [Admati & Pfleiderer \(1988\)](#)
- **Close**: index funds and ETFs trade at the closing price, which is fixed by the closing auction
- Practical rule: an execution algorithm follows the volume curve to reduce impact (Section 7)

Bitcoin Around the Clock

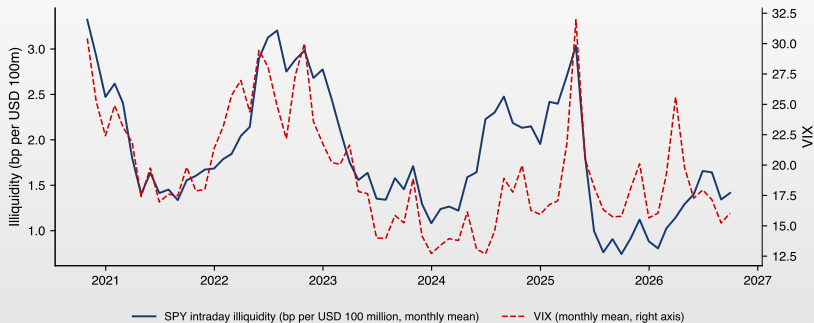


□ Mean absolute 5-minute return by hour (UTC), September 2024 – September 2026; 208 596 bars

Reading the Bitcoin Clock

- Weekdays: peak of 16.7 bp at 14:00 UTC (US open and macro releases), low of 7.3 bp at 10:00 UTC
- Weekend: 5.9 bp against 9.9 bp on weekdays, 1.69 times calmer; seminar B7 attaches day-bootstrap intervals to both contrasts
- A 24/7 market still follows the clock of banks, ETFs and traditional investors; high-frequency Bitcoin data also help forecast daily risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)

Liquidity and Volatility: SPY and the VIX



- Daily intraday illiquidity: mean of $|r|$ / traded value over the 78 five-minute bars, in bp per USD 100 million; monthly means

Reading Liquidity and Volatility

- ▣ Correlation of daily illiquidity with the VIX: 0.58; of monthly means: 0.69
- ▣ Regression of log illiquidity on log VIX: elasticity 0.95 (HAC standard error 0.12)
- ▣ April 2025: illiquidity 3.04 against a sample mean of 1.83
 - ▶ Liquidity providers widen quotes and shrink depth exactly when investors most need to trade
- ▣ Liquidity risk is therefore correlated with market risk [Acharya & Pedersen \(2005\)](#)

Recap: Illiquidity and Intraday Patterns

- ▣ The Amihud ratio separates markets by orders of magnitude: BVB blue chips show an absolute-return-to-traded-value ratio thousands of times larger than US large caps
- ▣ Volume and volatility are U-shaped over the trading day, with spikes at scheduled news
- ▣ Bitcoin, open 24/7, is most active when US markets are open and calmest at weekends
- ▣ Illiquidity rises with volatility: liquidity disappears when it is most needed

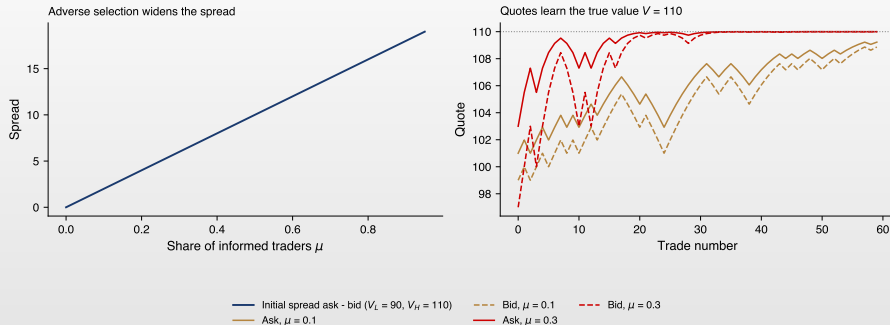
The Glosten–Milgrom Model

- Value $V \in \{V_L, V_H\}$, $P(V = V_H) = \theta$; each period one trader arrives [Glosten & Milgrom \(1985\)](#)
 - ▶ With probability μ informed (buys if $V = V_H$, sells if $V = V_L$), otherwise uninformed, trading for liquidity needs (buys or sells with probability $1/2$) [Bagehot \(1971\)](#)
- Competitive, risk-neutral market maker: zero expected profit on each trade
- $a = \mathbb{E}[V \mid \text{buy}]$, $b = \mathbb{E}[V \mid \text{sell}]$
- $P(\text{buy} \mid V_H) = \mu + \frac{1-\mu}{2}$, $P(\text{buy} \mid V_L) = \frac{1-\mu}{2}$
- After each trade, Bayes' rule updates θ : a buy raises it, a sell lowers it

Glosten–Milgrom: the Spread

- $P(V_H \mid \text{buy}) = \frac{\left(\mu + \frac{1-\mu}{2}\right) \theta}{\left(\mu + \frac{1-\mu}{2}\right) \theta + \frac{1-\mu}{2}(1-\theta)}, \quad a = V_L + (V_H - V_L) P(V_H \mid \text{buy})$
- At $\theta = 1/2$: $a - b = \mu(V_H - V_L)$
 - ▶ The spread grows linearly with the share of informed traders
 - ▶ $V_L = 90$, $V_H = 110$: spread 2 for $\mu = 0.1$, 6 for $\mu = 0.3$
- No inventory, no fees: the spread exists purely because of information
- Trade prices form a martingale: the market is efficient in the semi-strong sense (Chapter 2)

Glosten–Milgrom in Action



- Left: initial spread as a function of μ ; right: one simulated sequence of 60 trades with true value $V = 110$ (same random draws for both μ)

Reading the Learning Paths

- With $\mu = 0.3$ the bid exceeds 108 after 7 trades; with $\mu = 0.1$ only after 48
- More informed traders: wider spreads, but faster price discovery
 - ▶ The trade-off between liquidity and informational efficiency
- The spread narrows as the market maker learns: less remains to be learned from the next trade
- Empirical counterpart: the probability of informed trading (PIN) [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)

The Kyle Model

- One insider knows $v \sim N(p_0, \sigma_v^2)$ and submits a market order x [Kyle \(1985\)](#)
 - ▶ Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v
 - ▶ The market maker sees only the total order flow $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$
- The insider chooses x to maximise $\mathbb{E}[(v - p)x]$, knowing that his own order moves the price
- Linear equilibrium: $x = \beta(v - p_0)$, $p = p_0 + \lambda y$

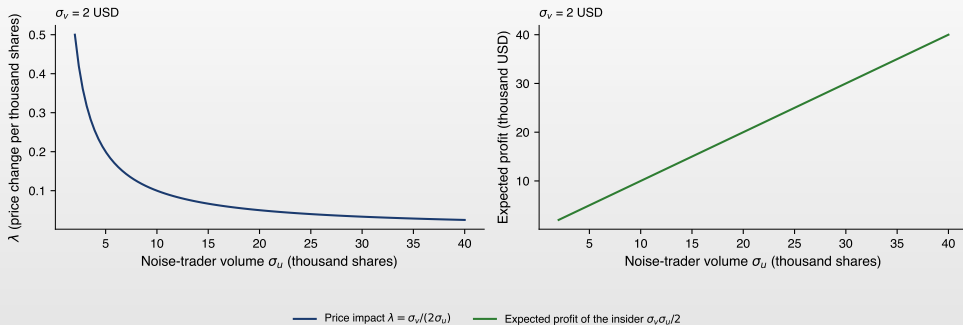
Deriving the Kyle Equilibrium

- **Insider:** given $p = p_0 + \lambda(x + u)$, maximise $\mathbb{E}[(v - p_0 - \lambda x - \lambda u)x \mid v] = (v - p_0)x - \lambda x^2$
 - ▶ First-order condition: $x = (v - p_0)/(2\lambda)$, so $\beta = 1/(2\lambda)$: he trades less aggressively because his own order moves the price
- **Market maker:** given $x = \beta(v - p_0)$, (v, y) is jointly Normal, so $\mathbb{E}[v \mid y]$ is the linear projection
 - ▶ $\lambda = \frac{\text{Cov}(v, y)}{\text{Var}(y)} = \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2 + \sigma_u^2}$ (the population OLS slope of v on y)
- Substitute $\beta = 1/(2\lambda)$: $\beta^2\sigma_v^2 = \sigma_u^2$, hence $\beta = \sigma_u/\sigma_v$ and $\lambda = \sigma_v/(2\sigma_u)$
- Posterior variance $\text{Var}(v \mid y) = \sigma_v^2 - \lambda^2\text{Var}(y) = \sigma_v^2 - \frac{\sigma_v^2}{4\sigma_u^2} 2\sigma_u^2 = \sigma_v^2/2$

Kyle: the Equilibrium

- $\beta = \frac{\sigma_u}{\sigma_v}, \quad \lambda = \frac{\sigma_v}{2\sigma_u}, \quad \mathbb{E}[\text{profit of the insider}] = \frac{\sigma_v \sigma_u}{2}$
- λ ("Kyle's lambda"): price move per unit of order flow; $1/\lambda$ is **market depth**
 - ▶ More noise trading (σ_u up): deeper market, larger insider trades, larger insider profit
- $\text{Var}(v \mid p) = \sigma_v^2/2$: the price reveals exactly half of the private information
- The insider hides behind noise traders: his order has the same variance as their orders, $\text{Var}(x) = \sigma_u^2$

Kyle's Lambda and the Insider's Profit



□ $\sigma_v = 2 \text{ USD}$; σ_u from 2 000 to 40 000 shares

From Models to Measurement

- Kyle's λ is estimated by regressing price changes on signed order flow; the Amihud ratio is its daily proxy [Hasbrouck \(2009\)](#)
- PIN: estimated from daily counts of buys and sells, by maximum likelihood [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)
- VPIN: order imbalance in volume time, proposed as a measure of flow "toxicity" [Easley, López de Prado & O'Hara \(2012\)](#)
 - ▶ Contested: correctly computed, VPIN did not predict the Flash Crash and is largely a transformation of volume and volatility [Andersen & Bondarenko \(2014\)](#)
- Order-flow imbalance at the best quotes explains most short-horizon price changes [Cont, Kukanov & Stoikov \(2014\)](#)

Estimating PIN, and Why It Is Fragile

- EKOP model [Easley, Kiefer, O'Hara & Paperman \(1996\)](#): each day an information event occurs with probability α (bad news with probability δ); given the day's information state, daily buy and sell counts B and S are independent Poisson
 - ▶ Rates: uninformed $\varepsilon_b, \varepsilon_s$; informed μ added to the buy side on good-news days, to the sell side on bad-news days
- $L = (1 - \alpha) \pi(B; \varepsilon_b) \pi(S; \varepsilon_s) + \alpha \delta \pi(B; \varepsilon_b) \pi(S; \mu + \varepsilon_s) + \alpha (1 - \delta) \pi(B; \mu + \varepsilon_b) \pi(S; \varepsilon_s)$, π the Poisson pmf;

$$\text{PIN} = \frac{\alpha \mu}{\alpha \mu + \varepsilon_b + \varepsilon_s}$$
- Numerics: for liquid stocks ($B \approx 10^4$) $e^{-\varepsilon} \varepsilon^B / B!$ under- or overflows; factor out common terms and use log-sum-exp [Lin & Ke \(2011\)](#)
 - ▶ Without it, estimates pile up at boundary values; the likelihood is also multimodal: use many starting values (seminar A9–A10)

The Hasbrouck Trade–Quote VAR

- VAR (vector autoregression) in the mid-quote return r_t and the signed trade x_t [Hasbrouck \(1991\)](#)
 - ▶ $r_t = \sum_{i \geq 1} a_i r_{t-i} + \sum_{i \geq 0} b_i x_{t-i} + \varepsilon_{1t}, \quad x_t = \sum_{i \geq 1} c_i r_{t-i} + \sum_{i \geq 1} d_i x_{t-i} + \varepsilon_{2t}$
- Identification: the trade comes first within the interval (b_0 enters, the reverse contemporaneous term does not): a recursive (Cholesky) ordering
- **Permanent impact:** the long-run cumulative response of the quote to a trade shock, $\lim_{h \rightarrow \infty} \sum_{k=0}^h \partial r_{t+k} / \partial \varepsilon_{2t}$: the information content of trades
- Variance decomposition of the efficient price: the share explained by trades generalises the adverse-selection share of MRR
- Like MRR, it needs signed trades; in Kyle's model the permanent impact is exactly λ

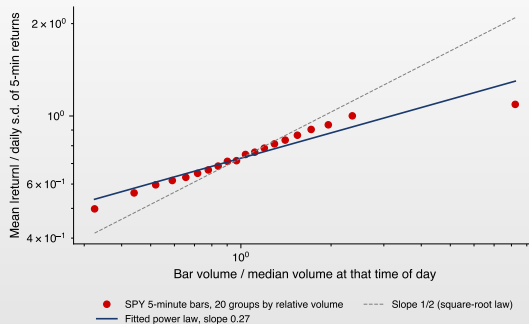
Order Flow Is Long-Memory and Self-Exciting

- Signs of market orders are autocorrelated with a power-law decay $\text{Corr}(\varepsilon_t, \varepsilon_{t+\ell}) \sim \ell^{-g}$, $0 < g < 1$: metaorders are split [Bouchaud, Gefen, Potters & Wyart \(2004\)](#)
 - ▶ Yet prices are close to a random walk: the propagator model $p_t = \sum_{s < t} G(t-s) \varepsilon_s + \text{noise}$ with a decaying kernel G reconciles the two
- Arrival times cluster: a Hawkes process has intensity $\lambda(t) = \mu + \int_{-\infty}^t \phi(t-s) dN_s$ [Bacry, Mastromatteo & Muzy \(2015\)](#)
 - ▶ Estimated by maximum likelihood; the branching ratio $\|\phi\|_1 = \int \phi$ is the share of events triggered by earlier events (endogeneity), stationary if < 1
- Links: durations and clustering of Chapter 8, intraday volatility of Chapter 9, the concave impact below

Metaorders and the Square-Root Law

- A **metaorder**: a large order split into many small trades over hours or days
- Empirical law for the peak impact (price move from the first to the last trade of the metaorder) [Tóth et al. \(2011\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#)
 - ▶ $\Delta P \approx Y \sigma_{\text{day}} \sqrt{Q/V_{\text{day}}}$, with Q the metaorder size, V_{day} the daily volume, Y of order 1
- Concave: trading 1% of daily volume moves the price by about 0.1σ , not 0.01σ : liquidity near the price is scarce and refills slowly (Section 2)
- Cost: if impact builds as $I(q) = Y \sigma \sqrt{q/V}$ during execution, the average price paid is $\frac{1}{Q} \int_0^Q I(q) dq = \frac{2}{3} I(Q)$ (the shortfall, before spread and fees); in currency, times Q and the price
- Testing it needs the traders' own order records; public bars allow only an indirect look (next slide)

Volume and Price Moves in SPY Bars



- 115 052 five-minute bars in 20 equal-size groups by relative volume (bar volume / median volume at that time of day)

Reading the Volume–Move Relation

- ▣ Mean $|r|$, scaled by the day's volatility, rises with relative volume with a slope of 0.27 on log-log axes
- ▣ Concave, as the square-root law predicts, but flatter than $1/2$
- ▣ Not a causal elasticity: **simultaneity**, news drives both volume and $|r|$; and opposite trades in a bar net out
 - ▶ An instrument for volume must move volume but not $|r|$ directly; seminar B6 tries the same bar's volume on the previous day and questions the exclusion restriction
- ▣ For execution costs use the metaorder law, calibrated on one's own trades

Optimal Execution: the Trade-Off

- Sell X shares within T days, in N steps of length $\tau = T/N$; holdings x_k with $x_0 = X$, $x_N = 0$; trades $n_k = x_{k-1} - x_k$
Almgren & Chriss (2001)
- Price dynamics: $S_k = S_{k-1} + \sigma\sqrt{\tau}\xi_k - \gamma n_k$; execution price $\tilde{S}_k = S_{k-1} - \eta n_k/\tau$; ξ_k i.i.d., mean 0, variance 1
 - ▶ γ : permanent impact; η : temporary impact, paid only by the trade itself
- **Trading fast**: high temporary impact, low price risk; **trading slowly**: the opposite
- Objective: minimise $\mathbb{E}[C] + \lambda \text{Var}[C]$ over deterministic schedules, with C the implementation cost and $\lambda \geq 0$ the risk aversion

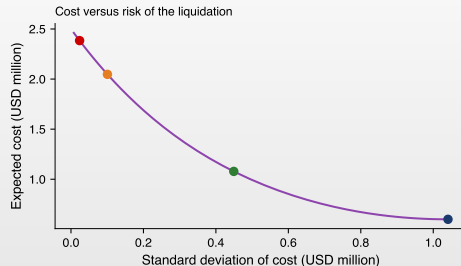
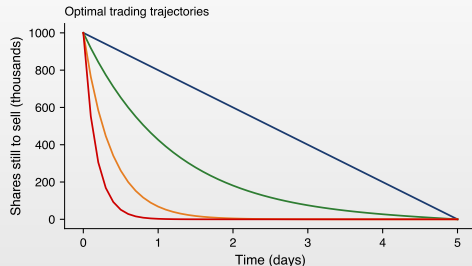
Deriving the Almgren–Chriss Trajectory

- Continuous time, holdings $x(t)$, trading rate $-\dot{x}$: temporary impact cost $\eta \int_0^T \dot{x}^2 dt$, price-risk variance $\sigma^2 \int_0^T x^2 dt$
 - ▶ Minimise $\int_0^T (\eta \dot{x}^2 + \lambda \sigma^2 x^2) dt$ subject to $x(0) = X$, $x(T) = 0$ (the permanent impact $\frac{1}{2}\gamma X^2$ does not depend on the path)
- Euler–Lagrange for $F(x, \dot{x}) = \eta \dot{x}^2 + \lambda \sigma^2 x^2$: $\frac{d}{dt} \partial F / \partial \dot{x} = \partial F / \partial x \Rightarrow \ddot{x} = \kappa^2 x$, $\kappa^2 = \lambda \sigma^2 / \eta$
- General solution $Ae^{\kappa t} + Be^{-\kappa t}$; the boundary conditions give $x(t) = X \sinh(\kappa(T - t)) / \sinh(\kappa T)$
- In discrete time κ^2 becomes $\frac{2}{\tau^2} (\cosh \kappa \tau - 1) = \lambda \sigma^2 / \tilde{\eta}$ (next slide); a resilient book with transient impact changes the shape [Obizhaeva & Wang \(2013\)](#)

Almgren–Chriss: the Optimal Trajectory

- $x_k = X \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)}, \quad \frac{2}{\tau^2} (\cosh(\kappa\tau) - 1) = \tilde{\kappa}^2 = \frac{\lambda\sigma^2}{\tilde{\eta}}, \quad \tilde{\eta} = \eta - \frac{\gamma\tau}{2}$
- $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum_{k=1}^N n_k^2, \quad \text{Var}[C] = \sigma^2\tau \sum_{k=1}^N x_k^2 \text{ (requires } \tilde{\eta} > 0)$
- $\lambda = 0$: $\kappa = 0$ and x_k falls linearly (trade evenly, the time-weighted schedule)
 - Large λ : most of the order is traded at the start, to cut the exposure to price risk
- $1/\kappa$ is the “half-life” of the trade: the time scale over which the position is worked off

Optimal Trajectories and the Efficient Frontier



— Risk-neutral, $\lambda = 0$ (sell evenly) — $\lambda = 2e-5$ — Efficient frontier (N = 5 days)
 — $\lambda = 2e-6$ — $\lambda = 1e-4$ (very risk averse)

- Numerical example of [Almgren & Chriss \(2001\)](#): $X = 10^6$ shares, $S_0 = 50$ USD, $T = 5$ days, $\sigma = 0.95$ USD per day^{1/2}, $\eta = 2.5 \cdot 10^{-6}$, $\gamma = 2.5 \cdot 10^{-7}$

Reading the Execution Chart

Strategy (5 daily steps)	Expected cost	Standard deviation
Even selling, $\lambda = 0$	0.60 million USD	1.04 million USD
$\lambda = 2 \cdot 10^{-6}$	1.08	0.45
$\lambda = 2 \cdot 10^{-5}$	2.05	0.10
$\lambda = 10^{-4}$	2.38	0.02

- With $\lambda = 2 \cdot 10^{-6}$ the first day sells 57% of the position; 428 599 shares remain
- Moving from even selling to $\lambda = 2 \cdot 10^{-6}$ cuts the standard deviation from 1.04 to 0.45 million USD, for 0.48 million USD more expected cost
- The frontier is convex: the first units of risk reduction are cheap, the last ones expensive

Execution Benchmarks

- ▣ **Implementation shortfall:** the difference between the paper portfolio at the decision price and the executed one [Perold \(1988\)](#)
- ▣ VWAP (Volume-Weighted Average Price): trade in proportion to the expected volume curve (the U-shape of Section 5)
- ▣ TWAP (Time-Weighted Average Price): equal amounts in each interval, the $\lambda = 0$ solution
- ▣ A participation algorithm (a fixed share of market volume) ignores price
 - ▶ This is the algorithm at the heart of the Flash Crash (next section)

Recap: Information, Impact and Execution

- ▣ Adverse selection alone creates a spread; Kyle's $\lambda = \sigma_v / (2\sigma_u)$ is a regression slope, estimable from order flow
- ▣ Impact is concave in size: the square-root law for metaorders
- ▣ Optimal execution trades off expected impact cost against price risk
- ▣ Risk-averse traders front-load; risk-neutral traders spread the order evenly

High-Frequency Trading: the New Market Makers



A microwave relay tower (Mojave, California); similar line-of-sight links carry trading signals between Chicago and New Jersey

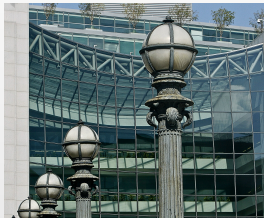
Photo: Tony Webster (2016); CC BY-SA 2.0; Wikimedia Commons

- HFT: algorithms that trade in micro- to milliseconds and end the day flat; speed from co-location and microwave links (faster through air than light in fibre)
- Low latency speeds up price discovery and narrows spreads [Hasbrouck & Saar \(2013\)](#)
- One HFT firm made markets on Chi-X and Euronext, mostly passively: it earned the spread and lost on positions [Menkveld \(2013\)](#)
 - ▶ Order flow and latency become “information” [O’Hara \(2015\)](#); market makers as intermediaries: [Pele \(2025\)](#)
- HFT liquidity comes and leaves fast: plentiful in calm markets, fragile under stress

Optimal Market Making: Avellaneda–Stoikov

- ▣ A dealer with inventory q maximises expected exponential (CARA) utility of terminal wealth, risk aversion γ ; mid-price s_t is a Brownian motion with volatility σ [Avellaneda & Stoikov \(2008\)](#)
 - ▶ Market orders hit a quote at distance δ from the mid with intensity $Ae^{-k\delta}$
- ▣ Reservation (indifference) price: $r = s - q\gamma\sigma^2(T - t)$: a long dealer ($q > 0$) values the asset below the mid
- ▣ Approximate optimal total spread around r (from an expansion of the value function, not the exact solution):
$$\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right)$$
- ▣ Inventory skews both quotes: this formalises the inventory component of the spread (Section 3); the $\ln(1 + \gamma/k)$ term prices the thinness of the order flow

The Flash Crash of 6 May 2010

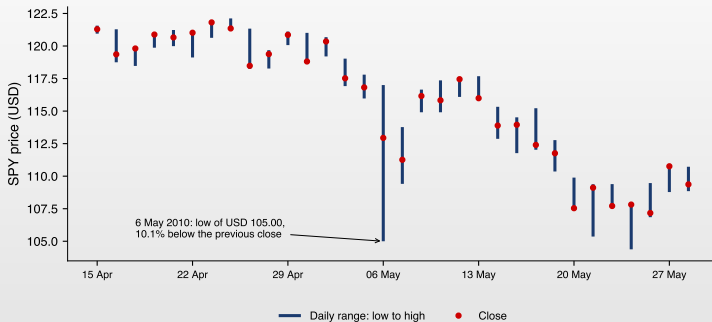


Headquarters of the US Securities and Exchange
Commission, Washington

Photo: David (dbking) (2008); CC BY 2.0; Wikimedia
[Commons](#)

- Report of the US futures (CFTC) and securities (SEC) regulators [CFTC & SEC \(2010\)](#)
- At 2:32 p.m. a mutual fund complex started selling 75 000 E-Mini S&P 500 futures (about USD 4.1 billion)
 - ▶ The algorithm targeted 9% of the previous minute's volume, "without regard to price or time"
- By 2:45:28 p.m. buy-side depth in the E-Mini had fallen below 1% of its morning level
- Over 20 000 trades in more than 300 securities were executed more than 60% away from their earlier prices, some at a penny

SPY Around the Flash Crash

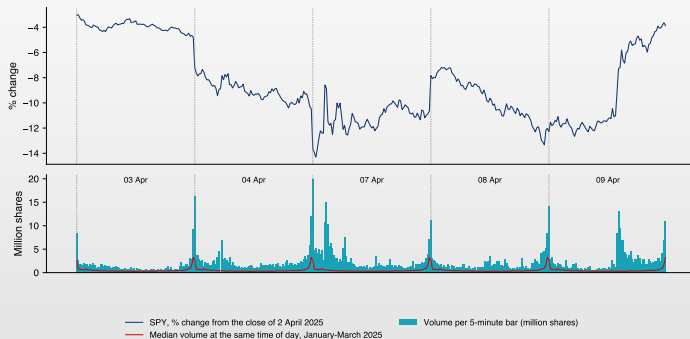


□ Daily low-high range and close, 15 April – 28 May 2010 (prices not adjusted)

Reading the Flash Crash

- ▣ SPY low of USD 105.00: 10.1% below the previous close of 116.82; it still closed only 3.3% lower
- ▣ Daily volume was 3.0 times the median of the previous three weeks
- ▣ HFT firms did not trigger the crash, but their reaction to the selling pressure amplified it [Kirilenko, Kyle, Samadi & Tuzun \(2017\)](#)
 - ▶ A 5-second pause in E-Mini trading at 2:45:28 p.m. let buyers return [CFTC & SEC \(2010\)](#)
- ▣ Responses: single-stock circuit breakers (a 5-minute pause after a 10% move in 5 minutes) and clearer rules for cancelling erroneous trades [CFTC & SEC \(2010\)](#)

A Recent Stress Episode: April 2025



- SPY, 5-minute bars, 3–9 April 2025; bottom panel: bar volume against the median volume at the same time of day in January–March 2025

Reading April 2025

- 7 April: intraday range of 8.6%; one 5-minute bar rose 4.28%, and a bar ten minutes later fell 2.92%
 - ▶ Realised volatility of the day: 6.7%, against a sample median of 0.61%
- 9 April: +10.0% from the open to the close
- Over the five days, volume was 4.6 times its usual level at the same times of day
- The bars document sharp moves, heavy trading and a high price impact per dollar (Section 5); they cannot establish price continuity, the absence of trading halts or order-book depth

The Speed Race and Frequent Batch Auctions

- In a continuous order book, correlations between related markets break down at millisecond horizons [Budish, Cramton & Shim \(2015\)](#)
 - ▶ This creates arbitrage opportunities that go to the fastest trader: an arms race in speed
- The cost of the race is borne by investors through wider spreads and thinner books
- Proposal: **frequent batch auctions**, e.g. every second, at a single price
- Competition moves from speed to price: narrower spreads and deeper markets in the model

Crypto Venues and Arbitrage

- 24/7 trading with no auctions; the same asset trades on many CEX order books at once, and on DEX pools priced by a formula (AMM, Chapter 16)
 - ▶ The same Bitcoin can trade at different prices on different exchanges [Makarov & Schoar \(2020\)](#)
- Price gaps are much larger **across countries** than within a country
 - ▶ Capital controls limit the movement of arbitrage capital
- A common component of signed volume across exchanges explains 80% of Bitcoin returns [Makarov & Schoar \(2020\)](#)
- Liquidity: high-low estimators track its time variation, Amihud-type ratios its level [Brauneis, Mestel, Riordan & Theissen \(2021\)](#); it can be forecast from market data [Agakishiev, Härdle & Pele \(2025\)](#)

Price Discovery across Venues: Information Shares

- Two prices of one asset, $y_t = (p_{1t}, p_{2t})'$, are $I(1)$ and cointegrated with $\beta = (1, -1)'$; VECM (vector error-correction model):
 - ▶ $\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^p \Gamma_i \Delta y_{t-i} + e_t$, $\text{Var}(e_t) = \Omega$ [Johansen \(1991\)](#)
 - ▶ Common efficient price: the stochastic trend with loadings $\psi' \propto \alpha'_\perp$, where $\alpha'_\perp \alpha = 0$
- **Hasbrouck information share**: venue j 's share of the variance of the common-trend innovation, $([\psi' F]_j)^2 / \psi' \Omega \psi$, F the Cholesky factor of Ω [Hasbrouck \(1995\)](#)
 - ▶ Depends on the ordering when Ω is not diagonal: report the bounds over both orderings
- **Gonzalo–Granger component share**: $\alpha_\perp$ normalised to sum 1; in $[0, 1]$ only if the two adjustment coefficients have opposite signs [Gonzalo & Granger \(1995\)](#)
- A price that does **not** adjust ($\alpha_j = 0$, weak exogeneity) leads price discovery; the inference is a Wald test of $\alpha_j = 0$, and non-rejection does not prove $\alpha_j = 0$

The Bucharest Stock Exchange



Offices of the Bucharest Stock Exchange

Photo: [Za.kurek \(2014\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- An electronic limit order book, as in the rest of this chapter [BVB \(history\)](#)
- Emerging-market status (FTSE Russell) since September 2020; Banca Transilvania and Nuclearelectrica were the first stocks in FTSE emerging-market indices
- A thin market: herding and bubbles in illiquid markets were studied on the BVB by [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
 - ▶ Bubbles and crashes are the subject of Chapter 17

BVB Liquidity: What the Data Say

- ▣ Amihud ratio of Banca Transilvania: 24.7 bp per USD 1 million; of Transelectrica: 1024.0; of SPY: 0.0016
- ▣ The BET ETF trades less than Banca Transilvania: 212.9 bp per USD 1 million
- ▣ Illiquidity has fallen since 2017, but the BVB remains thousands of times less liquid than US large caps
 - ▶ Daily spread estimates hide this: they rank BVB stocks with US stocks, because volatilities are similar
- ▣ Monthly market returns fall when illiquidity rises unexpectedly, on the BVB as in the US and in crypto (t-statistics -4.0, -3.8, -4.1; seminar Part C)

Price Discovery on the BVB: the BET ETF and the BET-TR Index

- ▣ Daily log closes of the Patria-TV BET ETF (it reinvests dividends) and the BET-TR index, 495 common trading days, September 2024 – September 2026; the basis has s.d. 0.84%; BET-TR is computed from its constituents' prices, not traded on a venue
- ▣ Johansen trace test: 49.6 for $r = 0$ (5% critical value 15.49), 0.10 for $r \leq 1$ (3.84): one cointegrating vector; VECM with 1 lag (BIC)
- ▣ Adjustment: ETF $\alpha = -0.12$ ($t = -2.11$); index $\alpha = -0.03$ ($t = -0.63$)
 - ▶ The index adjustment is not distinguishable from zero, the ETF's is: suggestive evidence that the ETF adjusts toward the index, with wide bootstrap intervals (seminar B8)
- ▣ Information share of the index: between 0.25 and 0.98; residual correlation 0.93. Component share of the index 1.39, outside $[0, 1]$ because both α are negative
- ▣ Lesson: with daily closes the innovations are almost collinear, so the bounds are wide; price discovery needs intraday data (seminar B8)

Recap: Stress, Crypto and the BVB

- ▣ HFT liquidity is cheap in normal times and fragile under stress (2010); price impact per dollar also rose in April 2025; market design can reduce the value of pure speed
- ▣ Cross-venue price discovery is a cointegration problem; daily data give wide information-share bounds
- ▣ Concentrated trading: two stocks carry more than half of the blue-chip traded value
- ▣ Use traded-value measures (Amihud), not daily spread proxies, to compare the BVB with other markets
- ▣ For portfolio decisions, the illiquidity of the BVB is a cost and a risk (Chapter 4 and seminar Part C)

Case Study: the Research Question

- **Question** Aquilina, Budish & O'Neill (2022): how much does **latency arbitrage**, the race to trade on public information, cost a whole stock market?
 - ▶ Latency arbitrage (sniping): trading against quotes made stale by a public price move before their provider can cancel them
 - ▶ Budish, Cramton & Shim (2015) modelled the race (Section 8) and measured one trade, futures–SPY arbitrage, at about USD 75 million a year; how to go from such trades to a whole market was unclear
- Why it was hard: order-book data record only the winners; a failed IOC order or a cancel that arrives too late leaves the book unchanged
 - ▶ **Message data**: every message to and from the exchange, including its “too late” replies, so both the winners and the losers of a race are seen
- Link to the chapter: effective spread = price impact + realised spread (Section 3)
 - ▶ Is sniping on **public** news a component of the cost of liquidity next to the **private**-information adverse selection of Glosten–Milgrom (Section 6)?

Case Study: Data and Method

- **Data** (Section II): London Stock Exchange message data for all FTSE 350 stocks, 17 August – 16 October 2015, 43 trading days, about 2.2 billion messages, obtained by the British regulator
 - ▶ Time stamps accurate to the microsecond (μs), taken where a message reaches the exchange's network, before any exchange processing
- **Race** (Section III): 2+ participants on the same stock, price and side; 1+ take and 1+ cancel attempts, or 2+ takes; at least one success and one failure
 - ▶ "Same time": within the **information horizon**, the first message's observed processing time plus the fastest reaction time seen ($29 \mu s$), capped at $500 \mu s$
- **Race profit** (Section IV.C): the signed gap between the race price and the mid-price 10 s later, times the displayed depth
- **Latency arbitrage tax** (Section IV.D): daily race profits / daily traded value, in bp; if races vanished, the cost of liquidity would fall by the non-race tax / non-race effective half-spread (eq. 7)
- **Extrapolation** (Section VI, Table XV): OLS of daily race profits on traded value V_t , and on V_t and 1-minute realised volatility σ_t times mean traded value $\bar{V}$, without a constant

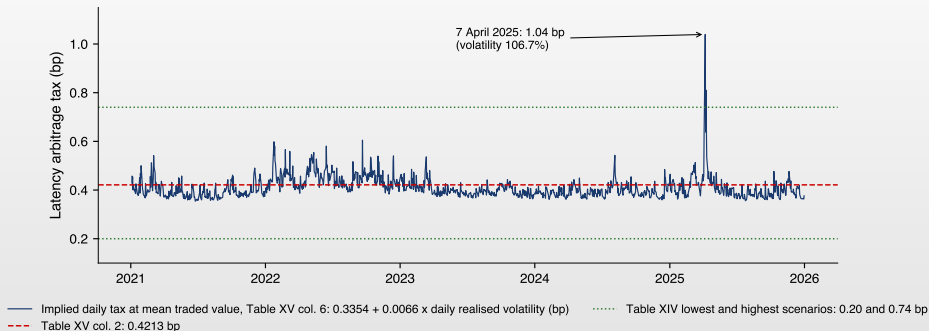
Case Study: Key Results

- **Frequent and fast** (Table II, Figure II): 537 races a day for the average FTSE 100 stock, about one a minute; the winner beats the first loser by 5–10 μs (mode), 46 μs (median)
 - ▶ 22.15% of FTSE 100 traded value is in races (Table III): speed decides a fifth of trading
- **Concentrated** (Figures III, IV): the top six firms win 82% of races and take about 80% of the liquidity taken in races, but provide only 42% of it
 - ▶ A race is mostly a transfer from slower liquidity providers to a few fast takers
- **Small prizes, large sums**: per FTSE 100 race, 0.48 ticks (1.20 bp) per share and £1.95 in total (Table VII); tax 0.419 bp of all traded value (Table X); $R^2 = 0.811$ of profits on volume (Table XV)
 - ▶ Global equities, 2018: USD 4.8 billion a year (range 2.3–8.4 across race definitions, Table XVII)
- **Cost of liquidity** (Tables XI, XIV): in the full sample, race price impact is 30.58% of all price impact and 32.82% of the effective spread; FTSE 100 realised spread -1.93 bp in races, $+0.15$ bp outside (one-way spreads, half the Section 3 measures)
 - ▶ In the paper's model, eliminating races (e.g. frequent batch auctions) would cut the cost of liquidity by 16.73% (Table XIII): sniping belongs next to adverse selection

Case Study: Replication on SPY

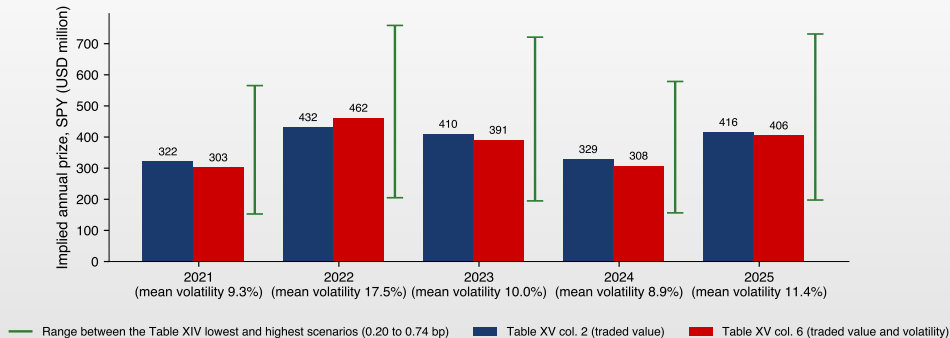
- **Replication** of the Section VI extrapolation for SPY (seminar C3): 5-minute bars from data/market, regular US session, the 247 full days of 2025
 - ▶ V_t = sum of close $\times$ volume, USD 9.88 trillion in 2025; σ_t from 5-minute log returns, annualised, mean 11.4%
 - ▶ Implied SPY prize in 2025: USD 416 million (Table XV, col. 2), USD 406 million (col. 6); USD 198–731 million in the lowest (0.20 bp) and highest (0.74 bp) scenarios of Table XIV
 - ▶ Col. 6 tax on a day with mean traded value ($V_t = \bar{V}$), $0.3354 + 0.0066 \sigma_t$ bp: 1.04 bp on 7 April 2025 ($\sigma_t = 106.7\%$), 2.5 times the average day
- **Two charts** (our computation, SPY, full days of 2021–2025)
 - ▶ The daily tax implied by col. 6 at mean traded value against the constant tax of col. 2 and the extreme scenarios of Table XIV
 - ▶ The annual prize implied by col. 2 and col. 6, year by year, with the range between the extreme scenarios

Case Study: the Implied Daily Tax on SPY



- Tax $0.3354 + 0.0066 \sigma_t$ bp (Table XV, col. 6) with σ_t from SPY 5-minute bars, 1 244 full days, 2021–2025 (our computation); $V_t = \bar{V}$, the paper's reading of col. 6 (Section VI); lines: col. 2 and the lowest and highest Table XIV scenarios, reference levels, not bounds on single days

Case Study: the Implied Annual Prize for SPY



- Annual prize $\sum_t \Pi_t$ in USD million under Table XV col. 2 and col. 6, with $\bar{V}$ the mean of each year; whiskers: 0.20 and 0.74 bp of traded value (lowest and highest scenarios, Table XIV); our computation, SPY, 2021–2025

Case Study: Reading the Replication

- **Daily tax:** median 0.40 bp, close to the 0.419 bp measured in London (Table X); above col. 2's 0.4213 bp on 30% of days
 - ▶ Peak 1.04 bp on 7 April 2025 ($\sigma_t = 106.7\%$); only 2 days, 7 and 9 April 2025, lie above the 0.74 bp of the highest scenario
 - ▶ The volatility term moves the tax only on stress days: most of the prize scales with traded value
- **Annual prize:** USD 322–432 million a year under col. 2; col. 6 is larger only in 2022 (462 against 432)
 - ▶ Col. 6 exceeds col. 2 when the mean volatility of the year is above $(0.4213 - 0.3354)/0.0066 = 13.0\%$; 2022 averaged 17.5%, the other years 8.9–11.4%
 - ▶ The choice of specification moves the prize by at most 7%; the lowest and highest Table XIV scenarios differ by a factor of 3.7: the tax level, not the model, is the main uncertainty
- For scale: the single futures–SPY trade of [Budish, Cramton & Shim \(2015\)](#) was about USD 75 million a year; the implied SPY prize is four to six times larger

Case Study: Limitations and an Open Extension

□ Limits stated by the authors

- ▶ One exchange, nine weeks of 2015; the extrapolation ignores differences in market structure (fragmentation, ETFs, geography) and other asset classes (Section VI.C)
- ▶ Races longer than the window, or whose losers see they are late and send nothing, are missed (Section V.A)

□ Limits of the replication

- ▶ The London coefficients are applied to SPY, not re-estimated: no US message data, so the charts show an implied prize, not a measured one
- ▶ Volatility is computed from 5-minute returns; the paper uses 1-minute returns

- **Open extension:** does SPY's daily EDGE spread (Section 3) co-move with the implied tax? Both depend on volatility, so this is only a diagnostic; identification needs an independent message-based race measure, a matched effective-spread benchmark and controls for volatility and traded value; a crypto exchange's message data (Section 9) would be a separate replication, not evidence on SPY

An Open Question: Do Price-Based Proxies Measure Trading Costs?

- Where trades and quotes are unavailable (BVB history, many crypto venues), costs are estimated from daily OHLC bars
 - ▶ Jahan-Parvar & Zikes (2023): low-frequency estimators track effective spreads only when the spread is large relative to volatility; otherwise volatility dominates
 - ▶ Ardia, Guidotti & Kroencke (2024): EDGE, an estimator from open, high, low and close prices, unbiased and efficient under their assumptions
- Formal question: $\hat{S}_i = S_i + b(\sigma_i) + \varepsilon_i$ across assets i
 - ▶ H_0 : $b \equiv 0$ (the proxy ranks assets by cost) against H_1 : b increasing in σ_i
 - ▶ Identification needs a cost benchmark measured independently of the proxy: effective spreads from matched trades and quotes; Amihud measures price impact, not the spread, and itself contains $|r|$
- Open: where direct cost data are scarce, is the spread-to-volatility ratio high enough for reliable estimation? Thin trading alone does not determine that ratio

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: collect validation studies, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Find studies that validate Roll, Corwin-Schultz, Abdi-Ranaldo and EDGE against effective spreads from trades and quotes; report the benchmark, the correlation in levels and in changes, the markets, and the DOI.
- 2. **Competing hypotheses**, each with a cross-sectional prediction
 - ▶ Give rival explanations for why Corwin-Schultz ranks BVB blue chips next to US large caps, and for each the predicted Spearman correlation of the estimate with volatility, with the Amihud ratio and with traded value.
- 3. **Data and code**: replicate the chapter's numbers first
 - ▶ Write Python that reads `data/market/<symbol>.csv` for the 25 assets of this chapter and computes Corwin-Schultz, with the overnight adjustment and negative two-day estimates set to zero. Target medians: US 53, BVB 54, crypto 124 bp

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: Spearman and partial Spearman correlations, bootstrap over assets, leave-one-market-out (US, BVB, crypto), a second window 2022-2024, and EDGE and Abdi-Ranaldo in place of Corwin-Schultz.
- 5. **Critique**: ask the assistant to attack the benchmark itself
 - ▶ Act as a referee: argue that the Amihud ratio is not a cost measure either (it scales with volume and with volatility); which benchmark would identify the spread, and what would it cost to obtain?
- 6. **Report**: every estimator and window, bootstrap intervals, the benchmark's limits, code and the AI log

What the Researcher Must Check

- **Tick-size benchmark:** no **quoted** spread is below one tick (effective spreads can be); compare every estimate with $0.01/P$ (0.21 bp for SPY) as a plausibility check, not a validation
- **Volatility contamination:** test: partial correlation with the benchmark controlling for volatility, and the 5-minute versus daily comparison of Section 3
- **Truncation:** negative estimates set to zero bias the mean upwards; report the share truncated and a median
- **Units and venues:** lei converted at the BNR rate; crypto volume summed over exchanges, some with inflated volume; test: rerun on a subset of venues
- **Small, clustered cross-section:** 25 assets in 3 markets are not exchangeable, and 3 clusters are too few for cluster-robust inference; report the asset bootstrap as conditional on independence, plus a within-market bootstrap and leave-one-market-out
- **Mistyped formulas:** check the constants of the estimator against the paper and run a unit test on simulated prices with a known spread

Mini-Case: Cost or Volatility?

- Replication: median Corwin–Schultz estimates US 53, BVB 54, crypto 124 bp (Section 3)
- A typical assistant proposal: “the estimates are similar, so BVB blue chips are about as liquid as US large caps”.
Diagnostic: compare the proxy’s rank correlation with volatility and with the Amihud price-impact ratio
- Test on the 25 assets, September 2024 – September 2026 (Spearman rank correlations)
 - ▶ With daily volatility: $\rho = 0.94$ ($p < 0.001$); with the Amihud ratio: $\rho = -0.12$ ($p = 0.58$)
 - ▶ Difference 1.06: 95% CI [0.58, 1.44] resampling assets as if independent, [0.71, 1.43] resampling within markets; partial ρ with Amihud given volatility: 0.27
- Interpretation: daily Corwin–Schultz ranks track volatility, not price impact, consistent with the contamination
[Jahan-Parvar & Zikes \(2023\)](#) predict for a low spread-to-volatility ratio; a diagnostic, not a test of H_0 , which needs matched effective spreads
- Code: `Quantlets/Ch_10/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: which estimator from daily bars, if any, measures the level of trading costs on the BVB and for crypto-assets?
- Replicate first: SPY Corwin–Schultz from daily bars 26.5 bp versus 3.11 bp from 5-minute bars and a tick floor of 0.21 bp (Section 3)
- Deliverables
 - ▶ A pre-registration (hypotheses, estimators, benchmark, tests, corrections) dated before the first run
 - ▶ EDGE implemented following [Ardia, Guidotti & Kroencke \(2024\)](#), validated on simulated prices with a known spread, then compared with the other estimators on the 25 assets
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`:
github.com/danpele/MFM/tree/main/project

Summary

- ▣ Prices are formed in limit order books: resting limit orders supply liquidity, market orders and marketable limit orders consume it
- ▣ The spread pays for processing, inventory and adverse selection; Glosten–Milgrom and Kyle model the last one
- ▣ Daily spread proxies are contaminated by volatility when the spread is small relative to volatility, as for the liquid assets here; the Amihud ratio separates markets by orders of magnitude
- ▣ Liquidity follows the clock (U-shape) and the VIX; it dries up under stress (2010) and price impact rises (April 2025)
- ▣ Impact is concave in size; optimal execution balances impact against price risk
- ▣ Inference matters: the Roll covariance is biased in small samples, structural models (MRR, Hasbrouck) need signed trades, price discovery is a VECM problem

Which Liquidity Measure When?

Available data	Suggested measure
Trades and quotes with timestamps	Effective and realised spread; MRR by GMM; Hasbrouck VAR; PIN
Prices of one asset on several venues	VECM; information-share bounds and component shares
Intraday bars (1–5 minutes)	Corwin–Schultz or Abdi–Rinaldo per day; intraday Amihud
Daily prices and volumes	Amihud ratio (price impact) across assets; EDGE for spread levels only if validated against effective spreads; high-low estimators mainly for changes over time
Own executions	Implementation shortfall; calibrate the square-root law
Theory, what-if analysis	Simulated order books; Glosten–Milgrom, Kyle, Almgren–Chriss

Coming Next: Continuous-Time Models (Chapter 11)

- ▣ From trades and order books back to smooth models of prices: Brownian motion and Itô calculus
- ▣ At high frequency, microstructure noise contaminates volatility estimates: realised volatility (Chapter 9) needs care
- ▣ Further reading: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Harris (2003), Franke, Härdle & Hafner (2019)

Further Study on Quantinar

Advanced courses on quantinar.com

 Applied Computational Finance

Code and data

 Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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